

A

Persona inputs

Profile sources used to condition the model



No profile baseline
empty prompt



Demographic survey profile
age, gender, education



Digital-twin survey summary
survey-derived person profile



Synthetic persona library
generated population profiles



Game-grounded persona generator
task-relevant axes



sample one profile for each scenario

B

Revealed-behavior matching

The model chooses among real observed trajectories

Sampled profile

e.g., dislikes free-riding,
values fairness, speaks rarely

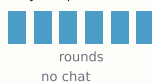


PGG transcript

players' contributions,
chat, rewards, punishments

Observed candidate behaviors

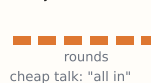
Player A
fully cooperative



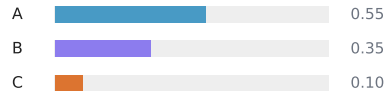
Player B
conditional cooperator



Player C
fully defective



LLM returns top-k matches with probabilities that sum to 1

**C**

Coverage and evaluation

Assess whether matched behavior spans the human support

Human behavioral support
all observed trajectories in the scenario



Matched distribution
probability-weighted selected trajectories

Behavioral checks

Behavioral skew

Do selected players over-contribute, over-reward, etc.?

Behavior-space coverage

Does the selected distribution cover observed behavior types?

Residual identity concentration

Does matching collapse after similar behaviors are grouped?

Support vs. sampling

Could reweighting personas recover the human distribution?